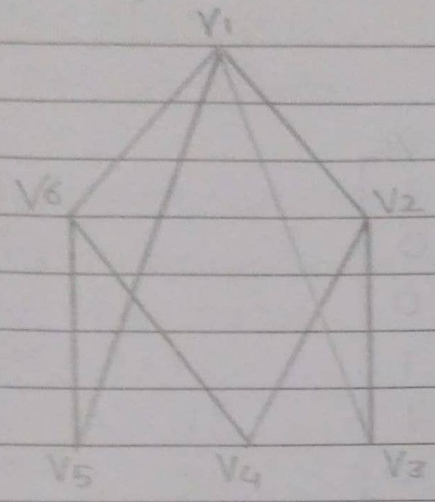
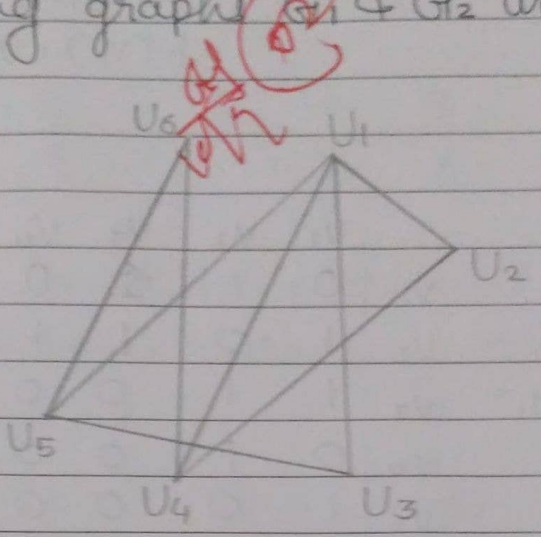
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Assignment -6

Q.1. Show that the following graphs G_1 & G_2 are isomorphic





Ans→

	Conditions	G_1	G_2
①	No of vertices	6	6
②	No of edges	8	8
③	degree sequence	4,3,2,2,2,3	4,2,2,3,3,2
④	one to one vertices	V1 V2 V3 V4 V5 V6	U1 U4 U2 U6 U3 U5
⑤	one to one edge.	F:(V1) = U1 f:(V2) = U4 f:(V3) = U2 f:(V4) = U5	f: $G_{11} \rightarrow G_{12}$ is isomorphic f^n & graph G_{11} is isomorphic to graph G_{12} $G_{11} \cong G_{12}$

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f	$f:(v_3) = u_3$ $f:(v_6) = u_5$

$MG_{G_1} =$

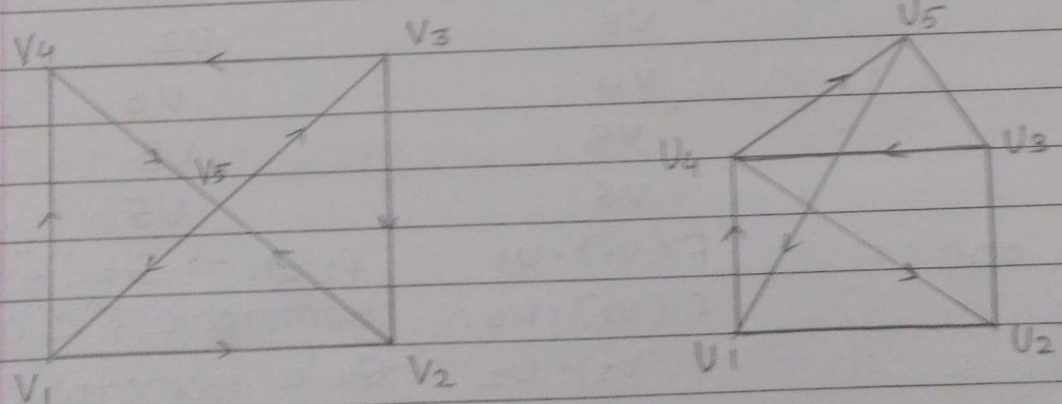
	v_1	v_2	v_3	v_4	v_5	v_6
v_1	0	1	1	0	1	1
v_2	1	0	1	1	0	0
v_3	1	1	0	0	0	0
v_4	0	1	0	0	0	1
v_5	1	0	0	0	0	1
v_6	1	0	0	1	1	0

$MG_{G_2} =$

	u_1	u_4	u_2	u_6	u_3	u_5
u_1	0	1	1	0	1	1
u_4	1	0	1	1	0	0
u_2	1	1	0	0	0	0
u_6	0	1	0	0	0	1
u_3	1	0	0	0	0	1
u_5	1	0	0	1	1	0

$F: V_1 \to V_2$ is isomorphic function and $V_1 \cong V_2$

Q.2



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Conditions	G1	G2
1] No of vertex	5	5
2] No of edges	8	8

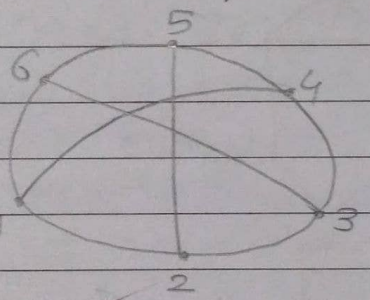
Indegree/outdegree sequence is preserved for G2
The corresponding indegree/outdegree pattern is the same.

f:(V1) = U1
f:(V2) = U2
f:(V3) = U3
f:(V4) = U5
f:(V5) = U4

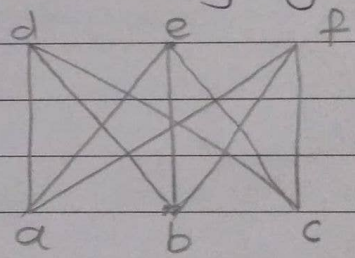
The direction of corresponding edges are preserved under mapping.

The Two graphs are isomorphic.

Q.3 Determine whether the following graphs G & H are isomorphic or not justify your answer



G



H.

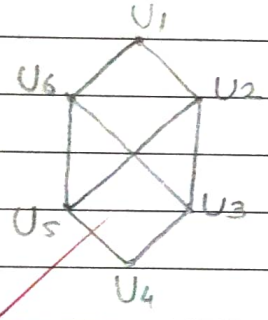
	G	H	
Vertices	6	6	both equal
edges	9	9	both equal
degree sequence	3,3,3,3,3,3	3,3,3,3,3,3	graph are regular

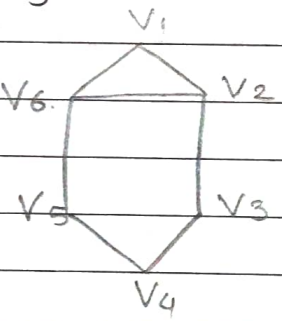
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one to one vertices	1 → b 2 → e 3 → f 4 → c 5 → d 6 → a	Validated
one to one mapping of edges	f:(1) ⇒ b f:(2) ⇒ e f:(3) ⇒ f f:(4) ⇒ c f:(5) ⇒ d f:(6) ⇒ a.	edges are also mapping one to one

$G \cong H$ The graphs are isomorphic.

Q.4. Determine graph G & H shown in figure are isomorphic or not justify your answer.





	G1	G2
No of vertices.	6	6
No of edges.	8	8
degree sequence	3,3,3,3,2,2	3,3,3,3,2,2
one to one mapping of vertices.	U1 U2 U3	V1 V2 V5

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	U_4	V_4
	U_5	V_3
	U_6	U_6
One to one mapping of edges	$f:(U_1) = V_1$	
	$f:(U_2) = V_2$	
	$f:(U_3) = V_5$	
	$f:(U_4) = V_4$	
	$f:(U_5) = V_3$	
	$f:(U_6) = V_6$	

$\therefore G_1 \cong H$.

Q.5 Decide which of the following graphs are eulerian or hamiltonian or both and write down as eulerian circuit and hamiltonian circuit wherever possible.

(a) The graph has 4 vertices & 6 edges degrees: $d(1)=3$, $d(2)=3$, $d(3)=3$, $d(4)=3$ since all four vertices have odd degree
not eulerian
But it has hamiltonian circuit.
$1 \to 2 \to 3 \to 4 \to 1$
→ Hamiltonian but not eulerian.

b) No of vertices = 5
No of edges = 10
degree of every vertex = 4
Since every vertex has even degree
[eulerian]

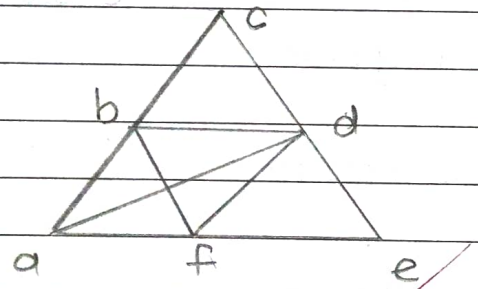
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1 → 2 → 3 → 1 → 4 → 2 → 5 → 3 → 4 → 5
all 10 edges are covered exactly once
Hamiltonian circuit:
1 → 2 → 3 → 4 → 5 → 1
Both eulerian & Hamiltonian.

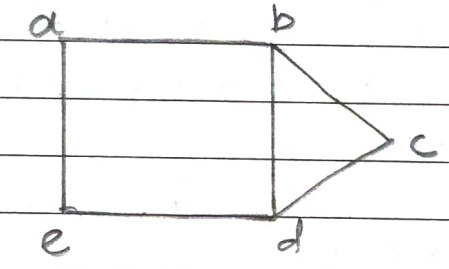
c) This is a cube graph.
No of vertices = 8
No of edges = 12
every vertex has degree 3.
since all vertices have odd degree
Not eulerian

d) Not eulerian or hamiltonian.

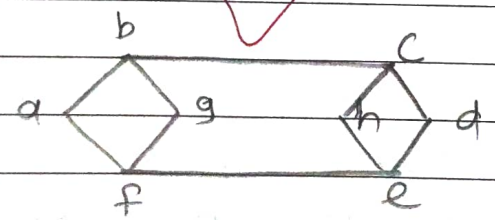
Q.6 Determine whether the given graph has a hamiltonian circuit or eulerian circuit if it doesn't find such a circuit



(i)



(ii)



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(i) degree of vertices.

$d(a)=3 \quad d(b)=4 \quad d(c)=2 \quad d(d)=5 \quad d(e)=2$

Since a & d have odd degree.
No ~~durian~~ eulerian circuit.
but a hamiltonian circuit exists.
$a \to b \to c \to d \to e \to f \to a$.
Hamitonion but not ~~eu~~ eulerian.

ii) Degrees.

$d(a)=2 \quad d(b)=3, \quad d(c)=2, d(d)=3, d(e)=2$

Since a & b have odd degrees.
No ~~et~~ eulerian circuit.
hamitonian circuit.
$a \to b \to c \to d \to e \to a$.
Hamiltonian but not eulerian.

iii) degrees.

$d(a)=2, d(b)=3, d(c)=3, d(d)=2, d(e)=3$
$d(f)=3, d(g)=2, d(h)=2.$

~~Thes~~ There are four odd degree vertices.
b, c, e, f.
No eulerian circuit.
A hamiltonion circuit is.
$a \to b \to c \to d \to e \to f \to g \to a$.
hamiltonion but not eulerian.

Student Undertaking for Ethical Academic Practice

Vidyalankar Institute of Technology, Mumbai

Assignment No: 6

Branch: INFT Vertical: _____

Semester: 3 Division: C

Subject Name: Engineering Mathematics III Subject Code:BSC05

I hereby declare that for the assignment / academic activity submitted by me, I confirm ONE of the following statements by ticking (O) the appropriate box:

☑ (1) I have prepared the assignment / academic activity entirely by myself.

D (2) I have completed the assignment / academic activity with academic guidance or discussion from seniors / friends / peers, while ensuring the work is my own.

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I understand that selecting options (4) or (5) indicates unethical academic practice and may attract academic penalties, including rejection of the assignment or disciplinary action, as per the rules of Vidyalankar Institute of Technology, Mumbai. I submit this declaration truthfully and accept full responsibility for the same.

Student Name: Bhoomi Dhumal Roll No.: 26101C2003

Signature: Bhoomi Date: 10/09/2026